ARUSHA TECHNICAL COLLEGE

JUNCTION OF MOSHI-ARUSHA AND NAIROBI ROADS

P.O. BOX 296, ARUSHA-TANZANIA

TELEPHONE: +255-27-2503040/2502076, FAX: +255-27-2548337

WEBSITE: , http://www.atc.ac.tz E-MAIL: rector@atc.ac.tz

ELECTRICAL ENGINEERING DEPARTMENT

ORDINARY DIPLOMA IN ELECTRICAL ENGINEERING

FINAL YEAR PROJECT– 2023/2024

PROJECT TITLE: AUTOMATIC FRUITS SORTER AND COUNTING MACHINE

PRESENTED BY:

NAME	ADM NO	EXM NO
ENEA ELISAFI LUKAS	21030712205	
ABDULRAZAK KAISI KAIMU	21030712001	

NTA LEVEL: 6

SUPERVISOR NAME: MR. ISACK NKOLA

2024

DECLARATION

We declare that the project report entitled "Automatic fruits sorter and counting machine" authenticated work carried out by Abdulrazak Kaisi & Enea Elisafi Lukas for the practical fulfillment of the award for Diploma in Electrical Engineering and this work has not been submitted for similar purpose anywhere else except to department of electrical engineering at Arusha Technical College (ATC).

Student Name: ENEA ELISAFI LUKAS

Signature: …………………………….

Date: ……………………

Student Name: ABDULRAZAK KAISI KAIMU

Signature: …………………………….

Date: ……………………

Supervisor Name: ISACK NKOLA

Signature: …………………….

Date: ……………………………

ABSTRACT

The abstruse nature of an automatic fruit sorter machine stems from the sophisticated combination of cutting-edge technologies required for its successful operation. At its core, the system relies on computer vision, a field of artificial intelligence, to analyze visual data obtained from the fruits. Complex algorithms are employed to recognize and differentiate fruits based on size, color, and shape, enabling the machine to make real-time decisions regarding the sorting process. Machine learning plays a crucial role in enhancing the sorter's accuracy over time by continuously refining its ability to identify different fruit varieties and defects.

In addition to the software complexities, the hardware components of the automatic fruit sorter machine contribute to its intricacy. Robotic arms or conveyor systems are often integrated into the design to facilitate the physical sorting of fruits based on the criteria determined by the algorithms. Calibration and synchronization of these mechanical components with the software algorithms demand a high level of engineering precision.

Moreover, the adaptability of the machine to handle a diverse range of fruits and the ability to maintain optimal performance under varying environmental conditions pose significant challenges. Engineers working on such projects must address issues related to the speed of the sorting process, ensuring that it is not only accurate but also efficient in handling large volumes of fruits.

Overall, the abstruseness of the automatic fruit sorter machine lies in the need for interdisciplinary expertise, combining advanced software development, artificial intelligence, robotics, and mechanical engineering to create a seamlessly integrated and highly efficient sorting system for agricultural applications

ACKNOWLEDGEMENT

We would like to express my sincere gratitude to Arusha Technical College for providing the conducive academic environment and resources that facilitated the successful completion of this project. Special appreciation goes to Mr. Isack Nkola, my project supervisor and Fadhili Malisa my project coordinator for their invaluable guidance, support, and encouragement throughout the research and development process.

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LIST OF ABBREVIATIONS

ATC	Arusha Technical College
AFSM	Automatic Fruit Sorter Machine
V	Voltage
F	Frequency
A	Ampere
DC	Direct Current
LCD	Liquid Crystal Display
mA	Milliamp ere

CHAPTER ONE

INTRODUCTION

1.1 Historical Background

The agricultural industry plays a crucial role in providing food for the growing global population. As technology continues to advance, there is an increasing demand for innovative solutions to enhance efficiency and productivity in agriculture. One area where automation can significantly contribute is in the sorting and grading of fruits. Traditional manual sorting methods are labor-intensive, time-consuming, and often prone to human error. To address these challenges, the development of an automated fruit sorter machine has been proposed

1.2 Problem Statement

In the agricultural sector, the sorting and grading of fruits are traditionally carried out through manual labor, a process that is not only time-consuming but also susceptible to inconsistencies and errors. As the demand for efficiency and precision in fruit processing increases, there is a critical need for the development of an Automated Fruit Sorter Machine. The current manual methods of sorting fruits in processing facilities pose several challenges that hinder the overall efficiency and quality of fruit production.

1.3 Objectives

These project objectives are divided into two parts, which are;

i. main objective,

ii. Specific objectives.

1.3.1 Main Objective.

The main objective is to design and "Automatic Fruit Sorter and Counting Machine" that can be used for sorting and counting fruits.

1.3.2 Specific Objectives

i. To design fruits color sorting system.

ii. To design fruits counting system.

iii. To design color detection system.

iv. To design conveyor system.

1.4 Scope of the Project

The project is aimed at providing an automatic fruit sorting process, that will help Agriculture personnel and Industrial goods producer to maintain process of sorting fruits. The scope of this project is aimed at fruits agriculture process, production industries based on fruits and in fruits Markets for monitoring of fruit sorting process.

CHAPTER TWO

LITERATURE REVIEW

2.1 General Introduction

An automatic fruit sorter and counting machine is a technologically advanced device designed to streamline the process of sorting and grading fruits based on various characteristics such as size and color. This automation in fruit sorting enhances efficiency, reduces labor costs, and ensures a consistent and accurate sorting process. The machine is widely used in the agricultural and food processing industries to handle large volumes of fruits quickly and efficient.

The sorting system performed by various systems, as follow

I. Conveyor System:

The conveyor belt is the primary component responsible for carrying and transporting materials. It is usually made of durable materials such as rubber, PVC, or metal, depending on the application and the nature of the transported items.

II. Vision System:

High-resolution sensors capture detailed of the fruits. These devices are strategically placed to provide comprehensive coverage of the sorting area. The quality of these sensors is vital for accurate fruit inspection.

III. Counting System:

The counting system relies on sensors or counting devices to detect specific events, items, or actions that need to be counted. These can include ultrasonic sensor, that will be used to detect the number of actions, also LCD will be used to display how many numbers of action that detected by the ultrasonic sensor and distinctly display them.

IV. Sorting Mechanism:

A sorting mechanism is a crucial component of systems designed to categorize and separate items based on specific criteria. In the context of an automatic fruit sorter machine, the sorting mechanism is responsible for physically directing fruits into different paths or bins according to their size, color, shape, or other predetermined characteristics

V. User Interface:

The machine is typically equipped with a user interface, which can be a graphical interface on a computer or a dedicated control panel. This allows operators to monitor the sorting process, adjust settings, and handle any exceptions or anomalies.

2.2 Background of the System

The historical background of automatic fruit sorter machines dates back to the early 20th century, marked by the introduction of basic sorting mechanisms relying on manual labor. As industrialization progressed in the 1930s and 1940s, more sophisticated technologies such as conveyor belts and basic mechanical devices improved the sorting process. The mid-20th century saw a significant leap with the integration of electronic sensors and computing technology for precise sorting. The 1980s and 1990s witnessed the widespread adoption of computer-based technologies. In recent decades, automatic fruit sorting machines have evolved with advanced sensors, computer vision, and machine learning, revolutionizing the fruit processing industry and minimizing the need for manual intervention in sorting processes.

2.3 Existing Systems

The term "existing system of an automatic fruit sorter machine" refers to the currently operational or established technology, design, and functionality of an automatic fruit sorter machine that is currently in use. This encompasses the hardware, software, and overall configuration of the machine as it functions within a specific context, such as a processing facility or agricultural operation.

2.2.1 TOMRA Sorting Solutions Genius

As of the last available information, the TOMRA Sorting Solutions Genius was a notable automatic fruit sorting machine introduced around 2005. The Genius series represented an advancement in sorting technology, particularly in the context of near-infrared (NIR) sorting capabilities. The machine incorporated sophisticated sensors and algorithms to analyze fruits based on their near-infrared spectral characteristics, enabling enhanced sorting accuracy.

Figure 1.Existing TOMRA Sorting Solutions Genius. Kim, H., et al, (2005)

2.2.2 Unitec - Cherry Vision 2

The Unitec - Cherry Vision 2 is an automatic fruit sorting machine designed specifically for cherries. Introduced around 2009, it represents an innovative solution for the efficient and precise sorting of cherries based on various criteria, including color, size, and external defects. Utilizing advanced vision technology, the Cherry Vision 2 employs high-resolution cameras and intelligent sorting algorithms to analyze each cherry in real-time as it moves through the system. This allows for accurate and reliable sorting, optimizing the quality control process and minimizing manual labor. The system's capabilities contribute to enhanced productivity and consistency in cherry processing operations, meeting the industry's demand for efficient and high-quality fruit sorting solutions.

Figure 2. Existing Unitec - Cherry Vision 2, Patel, R., Desai, P.,(2009)

2.2.3 Automatic fruit sorting

BBC Technology's Fresh LIFE represents a state-of-the-art automatic fruit sorting system that has significantly advanced the capabilities of fruit processing industries. fresh Life utilizes cutting-edge technology, particularly in the domain of computer vision, to enable high-speed and precise sorting of fruits based on various criteria such as size, color, and external defects. Introduced to the market with a focus on minimizing product damage and maximizing efficiency, Fresh life incorporates advanced sensors and intelligent algorithms to analyze and categorize fruits in real-time as they traverse through the sorting line.

Figure 3.Existing automatic fruit sorting Smith, J., Johnson, A (2022)

2.4: Block diagram of existing system

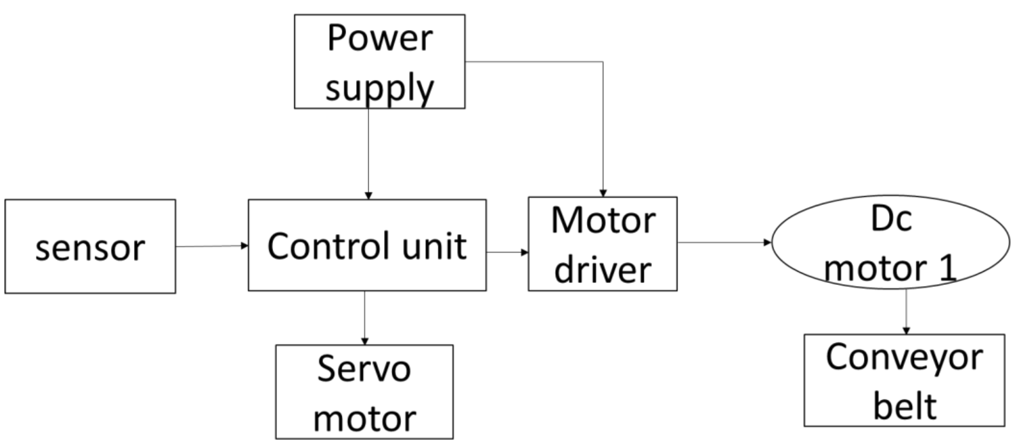

Figure 4.Block diagram of existing system

2.5 Disadvantages of the Existing system

I. Limited to count number of fruits.

II. Limited Adaptability to New Varieties.

III. Cost of Implementation.

IV. Dependency on Calibration.

2.6 Proposed System

"Introducing our Automatic Fruits Sorter Machine" play a solution for efficient fruit sorting. Equipped with ultrasonic sensors, it not only precisely categorizes fruits but also counts them in real-time. Revolutionize your agricultural processes with this smart, labor-saving technology for enhanced productivity and accuracy in fruit sorting.

A fruits sorter machine is a specialized piece of equipment designed to automate the process of sorting fruits based on various criteria such as size and color.

The proposed automatic fruit sorter machine incorporates an input conveyor system equipped with sensors for fruit detection, leading to an inspection unit with multiple sensors analyzing size, color, and other attributes. Data processing, utilizing machine learning algorithms, takes place at a central unit to determine sorting criteria, including size, color, and defects. An actuation mechanism, featuring pneumatic cylinders or robotic arms, diverts fruits into designated bins based on the sorting criteria. A user-friendly human-machine interface allows operators to monitor and control the process, while safety features, quality control mechanisms, and a rejection system ensure accuracy and reliability. Additionally, cleaning mechanisms and easy maintenance access contribute to the overall efficiency and hygiene of the system.

2.7 Block Diagram of Proposed System

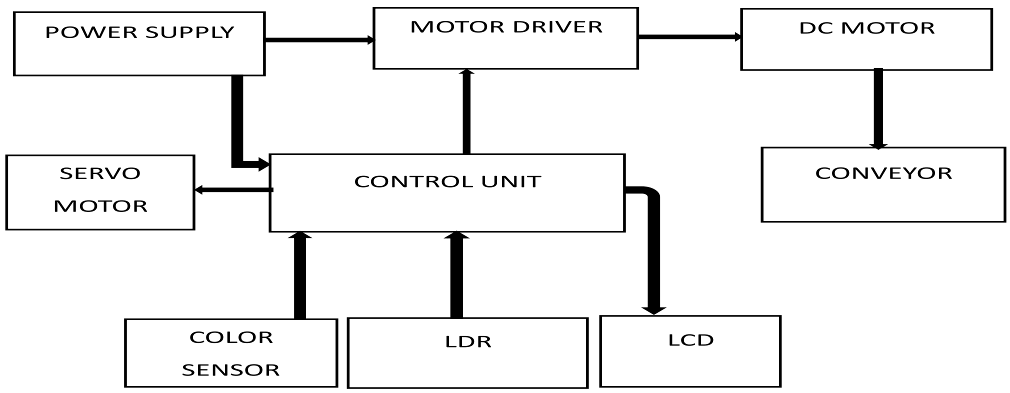

Figure 5.Block diagram of proposed system

2.8 Advantages of the Proposed Systems

i. Reduced Labor Costs: Automation minimizes the need for manual labor in sorting and counting tasks, leading to cost savings and allowing human resources to be allocated to more skilled and strategic roles.

ii. Flexibility: These machines can often be adjusted to handle various types and sizes of fruits, providing flexibility in sorting different produce with minimal reconfiguration.

iii. Real-time Information: Integration of display systems, such as LCDs, enables operators to access real-time information on the machine's performance, allowing for immediate adjustments if needed.

iv. High Speed Automatic fruit sorter and counting machines operate at a rapid pace, allowing for quick processing of a large quantity of fruits within a short timeframe.

v. Continuous Flow: These machines are designed to maintain a continuous flow of fruits through the sorting and counting process, minimizing downtime and maximizing productivity.

CHAPTER THREE

METHODOLOGY

3.1 General Introduction

Methodology shows various methods that will be used to ensure successful implementation of a fruit sorter machine. This section will explain in detail the steps that are to use in accomplishing the project. These steps and procedures include;

i. Literature Review

ii. Data Collection and Analysis

iii. Circuits Design and Testing

iv. Prototype Design and Testing

3.2 Literature review

In literature review various already existing fruit sorter machine will be comprehensively and critically analyzed. It will involves searching for evaluating, and summarizing relevant educational sources, such as books, journals, articles involving various ways of designing automatic fruit sorter machine.

3.3 Data Collection and Analysis

Data will be collected from various sources in the web, questionnaires will be used to ask various people working in agriculture and food processing industries how existing systems affect their day-to-day activities. The data collected will be analyzed by using google sheets and Mat lab to process various data collected from different sources, so as to implement the fruit sorter machine to be successful.

3.4 Circuit design and testing

The circuit will be designed by using proteus due to its vast compatibility with most of the electric components using in my project. It will be used to implement the circuit on software and test the working of all the key components.

3.5 Prototype designing and testing

The prototype to be designed is programmable. There are codes that are required to be developed and placed in the microcontroller (Arduino) using the Arduino IDE because of the use of Arduino. A program has to be placed in the control circuit and will help in all logic decisions and linking the input and output of the system. The physical system will then be tested to check if it operates according to its intended objectives.

CHAPTER FOUR

DATA COLLECTION, ANALYSIS AND PRESENTATION

4.0 Introduction

Data collection and analysis involve systematically gathering and examining data to make informed decisions and achieve project objectives. Data collection starts with defining objectives, identifying sources, selecting methods (e.g., surveys and questionnaire), and executing the process. Once collected, data undergoes cleaning to remove errors, exploration to understand patterns, and transformation for consistency. Analytical models are then built and trained on the data, with performance evaluated using metrics like accuracy or precision. Visualization tools help present insights clearly, and the process concludes with interpreting results and making data-driven recommendations. This structured approach ensures reliable, actionable outcomes for the project.

4.1 Primary data

These are the type of data which are collected directly from the application field. System primary data (user suggestions and Load ratings) were collected by interviewing different expected system users in the field area and also doing some observations. And answers I got from that field helped me to select some components in designing the project. The following are some examples of question which I asked in obtaining primary data for this project

i. Type of existing system used in fruit sorting? (Manual or automatic)
ii. Time used in making fruit sorting process?
iii. Speed required in making fruit sorting process?
iv. Challenges faced in using their system?
v. Torque required in driving shaft which hold the bobbin or forma?
vi. Between manual and automatic use which one is simple to use?
vii. Means used in counting number of fruits?

Due to those above question in designing an automatic fruit sorter machine, a questionnaire helps by gathering crucial insights from potential users and stakeholders about their needs, preferences, and expectations. It identifies the types of fruits to be sorted, desired sorting speed, accuracy requirements, and quality criteria such as size and color. Additionally, it assesses market demand, evaluates current solutions, and provides feedback on operational conditions, user interface preferences, safety concerns, and budget constraints. This information guides the design process, ensuring the machine is user-friendly, meets industry standards, and aligns with market needs, ultimately enhancing its effectiveness and commercial viability.

4.1.1 shows primary data which helps in designing the prototype which obtained through interview and questionnaires

pictorial analysis of questionnaire questioners received from different agriculture sectors and industries

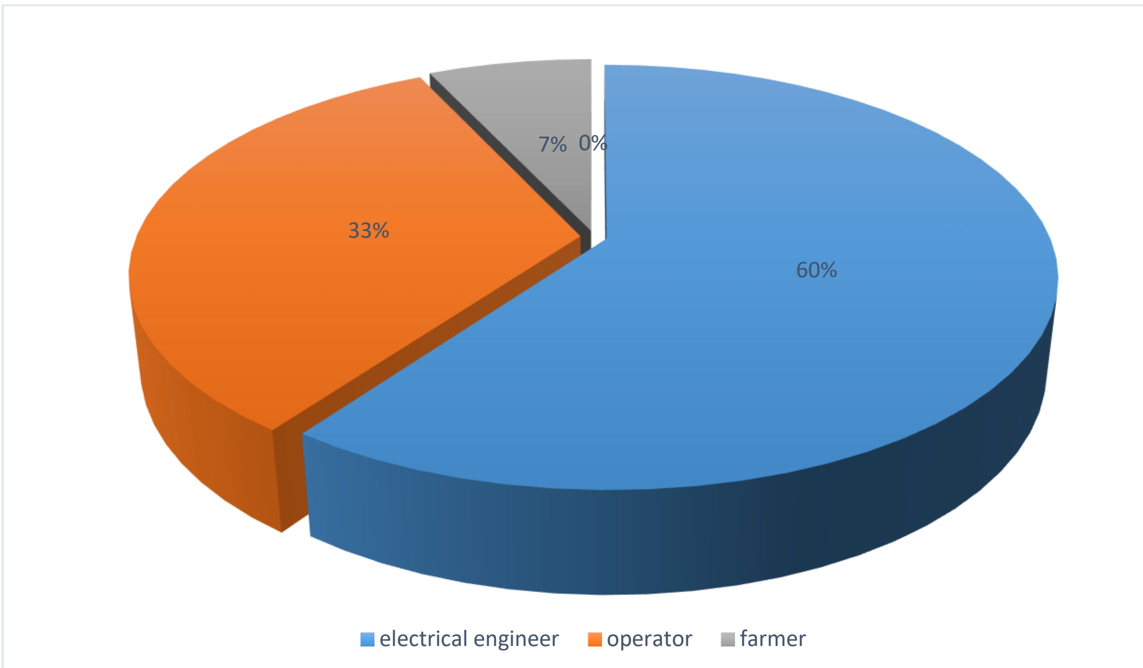

Figure 6.pictorial analysis of questionnaires received from different agriculture sectors and industries

4.2 Technical Data:

Mostly data collected where obtained by using questionnaire method, this done by giving multiple choice to people with deafness also in some institutions. The main areas include: installation of the device that will be used to solve the problem stated above. The following devices will be used to accomplish the project.

According to the technical data corrected the following data obtained.

4.2.1Power Supply

A DC power supply gives a constant DC voltage to the load irrespective of the load. Depending on its design it may be powered directly from a DC supply or an AC supply such as power mains. Basically, it is a device that converts an alternating current from utility supply to low regulated direct current is a DC power supply. Dc supply is used over AC supply simply because, most of modern electronics operate at very low voltages, and DC is used as low voltage.

Power supplies are critical components in electronic systems, providing the necessary voltage and current for proper operation. Here, we discuss the types of power supplies available in the market and analyse why a specific type is chosen for the automatic fruit sorter and counting machine.

S/N	**properties**	**Linear Power Supplies**	**Switching Mode Power Supplies (SMPS)**	**Battery Power Supplies**
1	Heat Dissipation:	Generate more heat and require larger heat sinks for thermal management.	Generate less heat due to higher efficiency, requiring smaller or fewer heat sinks.	Heat generated during charging/discharging, managed through thermal management systems.
2	Size and Weight	Larger and heavier because of bulky transformers and heat sinks	Compact and lightweight due to high-frequency components and smaller transformers.	Size and weight depend on battery type and capacity; can be relatively compact for portable applications.
3	Cost	Lower initial cost, but higher operational costs due to inefficiency	Higher initial cost due to complex design, but reduced operational costs due to high efficiency.	Variable cost; initial cost can be high for advanced chemistries, but operational cost is efficient.
4	Regulation and Stability	Good regulation but less efficient with large input variations	Excellent voltage regulation and can handle wide input voltage ranges.	Voltage regulation depends on BMS (Battery Management System); typically stable output.

5	Noise and Ripple	Lower noise and ripple, suitable for noise-sensitive application	Higher noise and ripple due to high-frequency switching, may require additional filtering.	Generally low noise and ripple, especially with appropriate regulation.

Table 1 Types of Power Supplies

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4.2.2 Control unit

The system includes a controller which will be controlling the input and output modules. For the system to a reliable and more efficient the most efficient and reliable control unit should be selected. The table below shows different controllers that may suit the system designing with their distinctive features and key parameters.

Specifications	AT89S52	PIC16690	ATmega16	ATmega328P
I/O pins	32	18	32	28
Operating voltage (V)	4.0-5.5	4.0-5.5	4.0-5.5	4.0-5.5
Operating Frequency	33MHz	20MHz	16MHz	16MHz
Architecture	RISC	RISC	RISC	RISC

Table 2 Types of Controllers

4.2.3Liquid Crystal Display (LCD).

This is a thin display device made of any colored or monochrome pixel array in front of the light or reflector. Each pixel consists of a column of liquid crystal molecules suspended between two transparent electrodes and two polarizing filters, the axes of polarity of which are perpendicular to each other. The liquid crystal twists the polarization of light entering one of the filters to allow it pass through the other. This helps the human being to interact with the common controllers being used in the devices or any operating circuit as it displays some of information which are due to program output (micro-controller output) or even simulation of the circuit. Some of the most commonly used LCDs are 16X1, 16X2, 20X4, and 128X16 Displays which means that 16 characters per line by 1 line, 16 characters per line by 2 lines, 20 characters per line by 4 lines, and 128 characters per line by 16 lines respectively. The LCD display offers the following advantages;

Specifications	LCD 16X1	LCD 16X2	LCD 20X1	LCD 128X16
Dimensions	80mm x 36mm	80mm x 36mm	96mm x 37mm	93mm x 70mm
Character	16	32	20	2048
Controller	HD44780	HD44780	HD44780	ST7920 or similar
power Consumption	low	medium	medium	Medium

Table 3 Types of Liquid Crystal Display(LCD)

4.2.4 Motors

A small DC motor is an electric motor that runs on direct current (DC) electricity and is typically used in applications requiring low power and compact size. These motors are commonly found in a variety of devices, including toys, small appliances, robotics, and other electronic devices; are as follow

DC Gear Motor:

A DC gear motor is a type of DC motor that is equipped with a gear assembly. The gear assembly helps reduce the motor's speed and increase its torque. These motors are used in applications where high torque at low speed is required.

Stepper Motor:

A stepper motor is a brushless DC electric motor that divides a full rotation into a number of equal steps. It is known for its ability to precisely control the position without the need for a feedback system.

Servo Motor:

A servo motor is a rotary actuator that allows for precise control of angular position, velocity, and acceleration. It typically consists of a motor, a position sensor, and a control circuit.

The below table show the difference in explained small DC motors

Specifications	Servo motor	Stepper motor	Dc geared motor
Operation	80mm x 36mm	80mm x 36mm	96mm x 37mm
Control	16	32	20
Torque	HD44780	HD44780	HD44780
Feedback	low	medium	medium

Table 4 Types of Motors

4.2.5 Motion sensor.

Motion sensors such as Passive Infrared (PIR), Ultrasonic, Microwave, Dual Technology, and Image Sensors employ varying technologies to detect movement: PIR senses infrared radiation, Ultrasonic emits and receives sound waves, Microwave detects reflected signals, Dual Technology integrates multiple methods, and Image Sensors capture visual changes. They are crucial in security, robotics, and automation for precise and reliable motion detection and monitoring.

Specification	**Passive Infrared sensor**	**Ultrasonic Sensor**	**Microwaves Sensor**	**Dual Technology Sensor**	**Image Sensor**
Detection principle	Infrared radiation detection	Ultrasonic wave reflection	Microwaves radiation reflection	Combination of two Sensor	Visual detection
Detection range	Short to medium Range	Short to medium Range	Medium to long Range	Medium to long Range	Medium to long Range
Detection angle	Wide		Wide	Wide	Wide
Response time	Very Fast	Fast	Fast	Moderate to Fast	Moderate to Fast
Accuracy	Moderate to High	High	High	High	High
Power consumption	Low	Low to moderate	Moderate to high	Moderate to high	High

Table 5 Types of Sensors

4.2.6 Resistors

Resistors are fundamental components in electronic circuits, designed to limit current flow. Types include Carbon Film, Metal Film, Wire wound, and Thick Film, each offering specific advantages like precision, stability, or high-power handling. Key specifications include resistance (ohms), power rating (watts), tolerance (% deviation from nominal value), and temperature coefficient (change in resistance with temperature). These properties ensure proper operation and reliability in diverse applications such as consumer electronics, telecommunications, and industrial equipment. The following are some types of resistors and their properties

Specification	Carbon film Resistor	Metal Film resistor	Wire wound Resistor	Thick Film Resistor
Material	Carbon	Metal	Metal wire	Cermet or metal oxide
Temperature coefficient	High (typically +_5% to +_10%)	Low High (typically +_1% to +_2%)	Low High (typically +_1% to +_5%)	Low High (typically +_1% to +_5%)
Power Rating	Low to Medium (typically 0.125W to 0.5W)	Medium to High (typically 0.25W to 2W)	High (typically 1W to 50 W)	Medium (typically 0.125W to 1W)
Tolerance	Wide (Typically +_5% to +_10%)	Narrow (Typically +_0.1% to +_1%)	Narrow (Typically +_1% to +_5%)	Medium to Narrow (Typically +_1% to +_5%)
Frequency stability	Moderate	High	Low	Moderate
Application	General purpose, Low cost	Precision, Stability	High power, precision	General purpose, cost effective

Table 6 Types of resistors

4.3 Data analysis.

For an automatic fruit sorter machine project, data analysis involves collecting high-resolution images and sensor data of various fruits, preprocessing the data to remove noise and normalize values, and labeling the data for accurate training. Feature extraction from images (color, size) and sensors (weight, size). Models are evaluated using metrics like accuracy and mean squared error, optimized through hyper parameter tuning, and deployed for real-time processing in the sorting machine. Continuous monitoring and periodic updates ensure the system maintains high accuracy and efficiency.

4.3.1 COMPONENTS SELECTION OF THE SYSTEM.

For this project to be accomplished, the following components together with their characteristics are explained below.

4.3.1.1Power Supply

A DC power supply gives a constant DC voltage to the load irrespective of the load. Depending on its design it may be powered directly from a DC supply or an AC supply such as power mains. Basically, it is a device that converts an alternating current from utility supply to low regulated direct current is a DC power supply. Dc supply is used over AC supply simply because, most of modern electronics operate at very low voltages, and DC is used as low voltage.

Power supplies are critical components in electronic systems, providing the necessary voltage and current for proper operation. Here, we discuss the types of power supplies available in the market and analyse why a specific type is chosen for the automatic fruit sorter and counting machine.

From the technical data obtained.

Chosen Power Supply: 12V DC Adapter

- **Specification:** 12V output, 2A current capacity
- **Type:** Switching Mode Power Supply (SMPS)
- **Purpose:** Provides a stable and efficient power source for the Automatic Fruits sorter and counting machine.

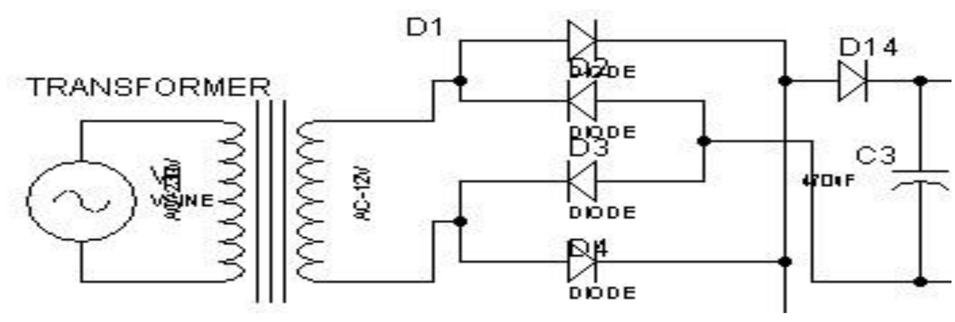

Figure 7.circuit diagram of DC power supply

From above circuit Transformer output determined on calculation

Transformer steps down Ac voltage from 220 -230VAC to 12VC.

Turns ratio (n)

I1N1=I2N2

I1/I2=V2/V1=N2/N1

V2/V1=12/230=0.052

Bridge rectifier calculation

Let

Vp (out) be Rectifier output

Vp (sec) be Transformer secondary voltage

Vp (sec) =1.414Vrms x 12˜16.96V

VP (out) = VP (sec) – 1.4v=16.96-1.4V=15.55Vpulsating

PIV for each diode = Vp (out) + 0.7V=16.25V

Vp (react) =Vp (sec) – 1.4V =15.55 – 1.4V = 14.15v

Taking ripple factor of 3.97%, then capacitor will be,

C=1/ (4fγR x (3)1/3) =1000uf, where γ=0.0397, f=50Hz, R=10K

Pure dc (filtered dc)

Vdc = (1-(1/2fRC)) x Vp (react)

Frequency for full wave rectified is 50Hz, let capacitor be 1000μF Rl be 10kΩ.

Vdc= (1-(1/100Hz x3.6kHz x1000Uf)) x14.15V = 12.18V

Part of the voltage is regulated by voltage regulator (L7805)

Up to 5V for microcontroller as well as LCD And is taken to feed up to other components.

Monitoring and Measurement

Reasons for Selection

1. **Efficiency:**

SMPS are known for their high efficiency, typically ranging from 85% to 90%. This efficiency is crucial for minimizing power losses and reducing heat generation within the system.

η=__Pin__×100%

Pout

For a 12V, 2A supply, assuming 90% efficiency:

$$Pin = \frac{24W}{0.90} = 26.67W$$

2. **Compact Size and Weight:**

SMPS units are typically more compact and lighter than linear power supplies, making them ideal for portable and space-constrained applications. This characteristic aligns with the project's goal of improving portability and design.

3. **Thermal Management:**

Due to their high efficiency, SMPS units generate less heat compared to linear power supplies, which is beneficial for maintaining the longevity and reliability of the system.Reduced heat generation correlates with better thermal management and lower cooling requirements.

4. **Stable Output Voltage:**

SMPS units provide a stable output voltage with minimal ripple, essential for the reliable operation of sensitive electronic components such as microcontrollers and sensors.

5. **Cost-Effectiveness:**

SMPS units are generally more cost-effective over the long term due to their higher efficiency and lower energy consumption. The initial cost might be higher, but operational savings and reduced thermal management costs justify the investment.

Calculation for Power Dissipation:

- Given the 12V, 2A SMPS powering the system, the total power provided is:

$$P.\ total = V \times I = 12V \times 2A = 24W$$

- The microcontroller (ATmega328P) and other components operate at 5V regulated by a 7805voltage regulator:

$$\text{Power dissipation} = (Vin - Vout) \times I$$

$$= (12V - 5V \times 1.5A$$

$$= 10.5W$$

Capacitors (10µF and 1µF)

Capacitors from reputable manufacturers are chosen for their quality and reliability in filtering out high-frequency noise and providing stable DC voltage to sensitive components like the microcontroller and sensors.

4.3.1.2 Control Unit

Microcontrollers are integral components in electronic systems, responsible for managing operations and interfacing with various peripherals. Here, is the discussion of the types of microcontrollers available in the market and analyse why the ATmega328P was chosen for the Automatic fruit sorter and counting machine

Selection of Microcontroller for the Project

Chosen Microcontroller: ATmega328P

- **Specification:** 8-bit AVR MCU, 32KB flash memory, 2KB SRAM, 1KB EEPROM, 20MHz clock speed.
- **Purpose:** Controls the Automatic fruit sorter and counting machine system, including sensor reading, counting, color detection and user interface management.

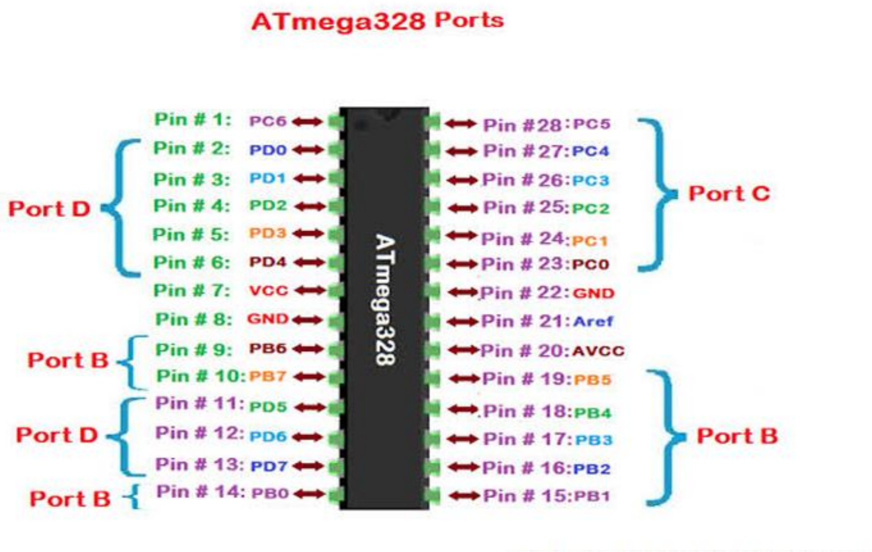

Figure 8.ATmega328P

Reasons for Selection:

1. **Adequate Performance for Project Needs:**

The ATmega328P provides sufficient computational power for the tasks required in the , reading Automatic fruit sorter and counting machine ,sensor inputs, and managing user interface elements.

For simple control tasks, an 8-bit microcontroller like the ATmega328P operates efficiently without unnecessary complexity

$$\text{Instruction execution time} = \frac{20MHz}{1} = 50ns \text{ per cycle}$$

The typical instruction execution in an AVR architecture is 1-2 cycles, which is adequate for the control tasks in this project.

2. **Low Power Consumption:**

The ATmega328P is designed for low-power operation, making it suitable for applications where energy efficiency is important.

Lower power consumption extends the life of battery-operated devices and reduces heat generation.

$$P = V \times I = 5V \times 0.02A = 0.1W$$

3. **Ease of Programming and Development:**

The ATmega328P is widely supported by development environments such as Arduino IDE, making it accessible for rapid development and prototyping.

This reduces development time and complexity, allowing for faster iteration and testing.

4. **Sufficient I/O Pins and Peripherals:**

With 23 programmable I/O lines, ADC channels, PWM outputs, and serial communication interfaces (UART, SPI, I2C), the ATmega328P can interface with various components and sensors required in the fume extractor.

Having multiple communication protocols and I/O options provides flexibility in system design.

Example:

- ADC for sensor inputs (e.g., air quality sensors).
- UART for serial communication with a user interface or debugging tools.

5. **Cost-Effectiveness:**

The ATmega328P is cost-effective(10000Tsh), balancing performance with affordability, making it ideal for budget-sensitive projects. The overall project cost is kept low without sacrificing the necessary features and capabilities.

Capacitors and Crystal Oscillator

22pF Capacitors and 16MHz Crystal Oscillator Capacitors Stabilizes the crystal oscillator. Crystal Oscillator has Specification: of and 16MHz Purpose Provides a stable clock signal for precise timing. Accurate timing is essential for reliable microcontroller operation, ensuring consistent performance across various tasks.

frequency stability:

fosc=16MHz

4.3.1.3Voltage regulator

This is a device that takes a variable or unstable input voltage and converts to a higher or lower constant output voltage that matches the needs of electronic circuits Since the power from the supply mains varies over time therefore rectified DC output is not stable due to voltage fluctuations hence voltage regulator required to stabilize it. (Components101, 2017) In this project, LM7805 Voltage regulator shown on the Figure below is used. It is a common voltage regulator in electronic circuits. The name 7805 signifies two meanings 78 means that it is a positive voltage regulator and 05 means that it provides 5V as output. Features and other technical details are found at the datasheet on Appendix B.

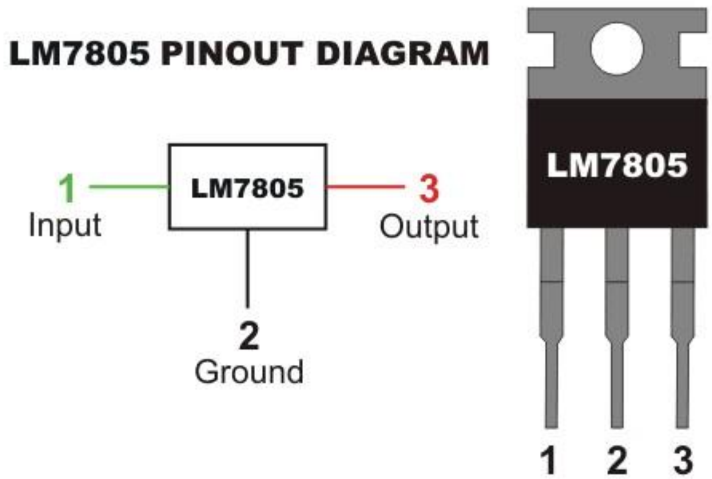

Figure 9.Voltage regulator

4.3.1.4Liquid Crystal Display (LCD).

This is the thin display device made of any coloured or monochrome pixel array in front of the light or reflector. Each pixel consists a column of liquid crystal molecules suspended between two transparent electrode and two polarizing filters, the axes of polarity of which are perpendicular to each other. The liquid crystal twists the polarization of light entering one of the filters to allow it pass through the other. This helps the human being to interact with the common controllers being used in the devices or any operating circuit as it displays some of information which are due to program output (micro-controller output) or even simulation of the circuit.

Advantages of LCD

1. Allow the display of the message.

2. It is cheap compared to a light-emitting diode display LED.

3. Easily programmable.

4. Use a small amount of power compared to LED.

5. Allow display of more than one content.

LCD with 16 characters per line by 2 lines (16 X 2) will be used, so as to obtain wider display of information**.** Figure below shows the LCD of 16 X 2 type.

In this project LCD is going to be used as the display

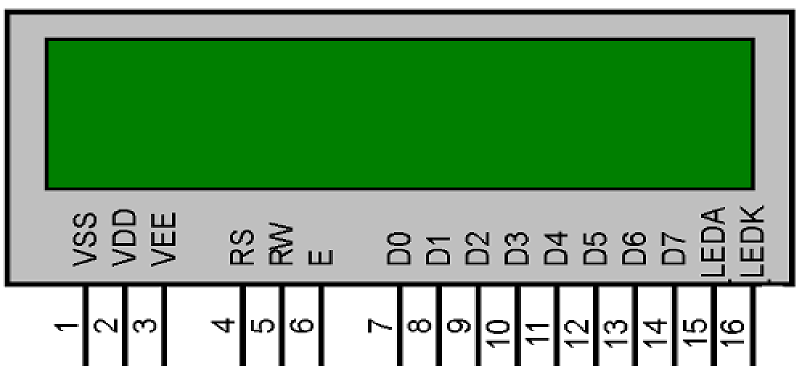

Figure 11.Liquid Crystal Display (LCD).

4.3.1.5Resistor

A device that has electrical resistance and that is used and that is used in electricity circuit for protection, operation or current control

Figure 12.resistor

Reasons for using resistor:

1. Current Limiting: Resistors control the flow of electric current to prevent damage to components by limiting excessive current.
2. Voltage Division: They divide voltage within a circuit, ensuring that components receive the appropriate voltage levels.
3. Biasing: Resistors provide the necessary biasing for transistors and other active components, ensuring they operate correctly.
4. Signal Conditioning: They are used in filtering and conditioning signals, such as in RC (resistor-capacitor) filters.
5. Pull-up/Pull-down Resistors: These ensure a known state for input pins in digital circuits, preventing floating inputs and reducing noise.

4.3.1.6 Color sensor

A color sensor is a device that detects and measures the color of an object by identifying the intensities of light in the red, green, and blue (RGB) spectrum. It typically uses photodiodes or phototransistors to convert light into electrical signals, which are then processed to determine the color. These sensors are used in various applications, including industrial automation, robotics, and consumer electronics.

Figure 13.Color sensor.

.

PIN NAME	PIN NUMBER	DESCRIPTION
GND	4	Power supply ground. All voltage are reference to the ground.
VCC	5	Supply voltage

OE	3	Enable for FO (Active low)
OUT	6	Output frequency (fo)
S0,S1	1,2	Select lines for output frequency scaling
S2,S3	7,8	Select lines for photodiode type
S0	S1	Output frequency scaling(fo)

Table 7 Specification of color sensor

Spesfication of color sensor

i. Single supply operation (2.7V – 5.5V)
ii. High resolution convertion of light intensity to frequecy
iii. Programable color and full scale output frequecy
iv. Power down feature
v. Communicates direct to microcontroller
vi. Size 28.4 * 28.4 mm

Reasons for using color sensor.

a. Accurate Color Detection: Color sensors can precisely identify and differentiate between various colors, ensuring accurate sorting of items based on their color.
b. High Speed: Color sensors can quickly process color information, allowing for fast and efficient sorting operations.
c. Automation: They enable automated sorting processes, reducing the need for manual intervention and increasing productivity.
d. Consistency: Color sensors provide consistent color detection and sorting, improving the reliability and quality of the sorted products.
e. Versatility: They can be used to sort a wide range of items, from fruits and vegetables to manufactured goods, making them highly versatile in various industries.

4.3.1.7 LDR (Light Depend Resistor).

A photo resistor (also known as a Photocell, or light-dependent resistor, LDR, or photo-conductive cell) is a passive component that decreases resistance with respect to receiving luminosity (light) on the component's sensitive surface Parameter Example Figures.

Max voltage @ 0 lux 200V

Peak wavelength 600nm

Min. resistance @ 10lux 1.8K ohms.

Figure 14.LDR (Light Depend Resistor).

Specifications:

- Operating Voltage: 5V DC
- Operating Current: 15mA
- Frequency: 40 kHz
- Measuring Angle: 15 degrees
- Resolution: 0.3cm

Purpose:

Reasons for selection LDR (Light Depend Resistor).

Light Sensing. Measures ambient light levels for automatic control of lighting.

Cost-Effective: Inexpensive sensor solution for light-dependent applications.

Analog Output. Provides analog voltage signals proportional to light intensity.

Simple Interface. Easy to integrate with microcontrollers for automation projects.

Reliability. Provides consistent performance in various lighting conditions.

4.3.1.8 Stepper motor

This is the electromechanical device that convert electrical pulse into precise mechanical movement. Unlike traditional motors which rotates continuosly when voltage is applied , stepper motor move in discrete steps or increment.

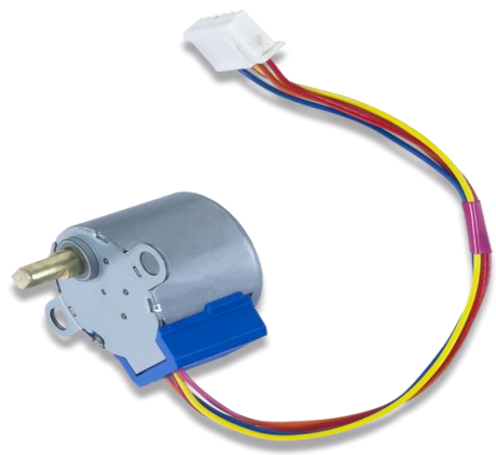

Figure 15.Stepper motor.

Rated voltage	5VDC
Number phase	4
Current	40mA
DC resistance	54 ohms
Phase inductance	3Mh
Frequecy	100Hz
Speed variation ratio	1/64
Stride angle	5.625 degree/64

Table 8 Specifications of stepper motor

Reasons for using stepper motor

a. Precise Position Control: Stepper motors can move to exact positions without the need for feedback systems, making them ideal for applications requiring precise positioning.
b. Repeatability: They provide consistent and repeatable movements, which is essential for tasks that require high accuracy.
c. Simple Control: Stepper motors are easy to control with digital signals, making them straightforward to implement with microcontrollers and other digital systems.
d. Full Torque at Standstill: They can hold their position with full torque when stopped, which is useful for applications where maintaining a position is crucial.
e. Open-Loop Control: Stepper motors do not require complex feedback systems, simplifying the control system and reducing costs.
f. Low-Speed Performance: They perform well at low speeds with smooth operation, which is beneficial for precise control application

4.3.1.9 Servo motor

Is the rotary or liner actuator that allows for precise control of angular or liner position , velocity and accerelation in mechanical system.

Figure 16.Servo motor .

Weight	9g
Rotation range	180 degree
Operating Temperature	-30 degree centigrade to +60 degree centigrade
Operating voltage	4.8v – 6v
Speed	0.08 sec/60 degree (6v)
Stall torque	1.8kg/cm (4.8v), 2.2kg/cm (6v)
Control system	PWM(pulse width modulation)

Table 9 Specification of servo motor

Reasons for using stepper motor

a. Precise Position Control: Servo motors provide accurate control of angular or linear position, which is essential for applications requiring exact movements.
b. High Torque at Low Speeds: They deliver significant torque even at low speeds, making them suitable for applications needing substantial force.
c. Closed-Loop Control: Servo motors use feedback systems (e.g., encoders) to constantly monitor and adjust position, speed, and torque, ensuring high precision and stability.
d. Smooth Operation: They offer smooth and consistent motion, which is crucial for applications like robotics, camera stabilization, and CNC machinery.

4.3.2 Simulation and results

Testing and quality control are an integrated part of any project process, Results of the project reflect to the objectives of the project.

Testing of the project is done by using the allowable power. The proposed circuit it designed to use (5 up to 12) VDC. If the allowable injected to circuit the result will be rotating of the motor according to the input of number of turns.

This chapter describes about the simulation of the designed system. It includes descriptions about simulation tools used, simulation constraints, performance testing parameters and simulation testing procedures. It also presents the results and their discussions.

Simulation Software

The simulation software for this project is proteus. The proteus simulation tool is chosen because of the following reasons;

a) Is rich in libraries of different types of electronic components and modules

b) It performs well in different varieties of projects ranging from simple electronic circuit to complex ones

c) In addition to simulation excellence, it offers a capability to prepare even the printed circuit board for actual Its realization of the prototypes of different designed systems.

CHAPTER FIVE

CIRCUIT DESIGN AND PROTOTYPE IMPLEMENTATION

5.1 Circuit Design

In this section, we review the processes that were included in the designing of the prototype of the project. For the design of the prototype and simulation a software program was used and the program was. The detailed design of the prototype is displayed below.

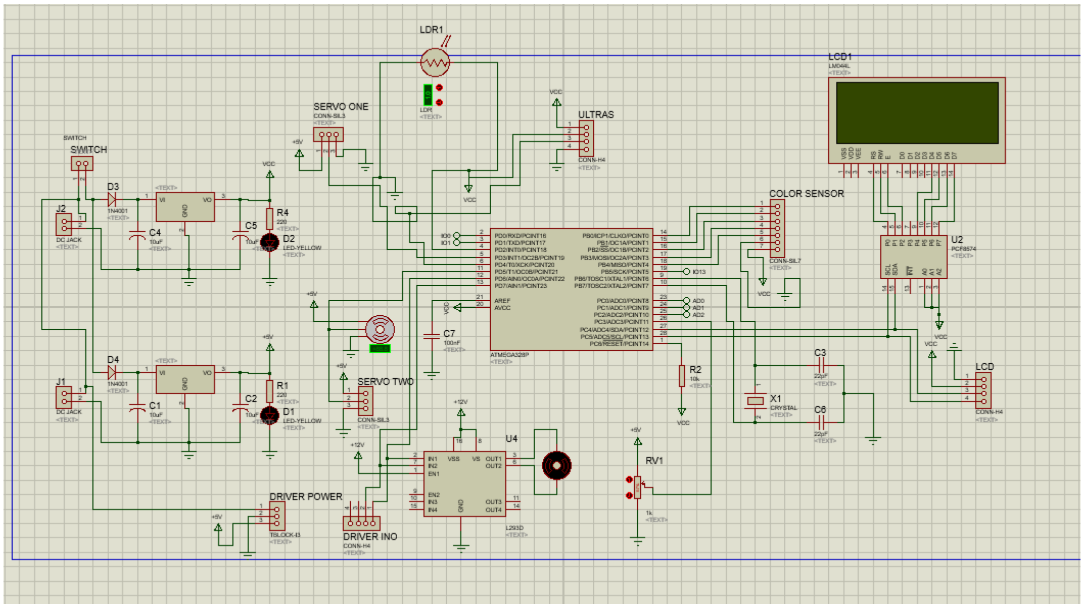

Figure 17. circuit simulation

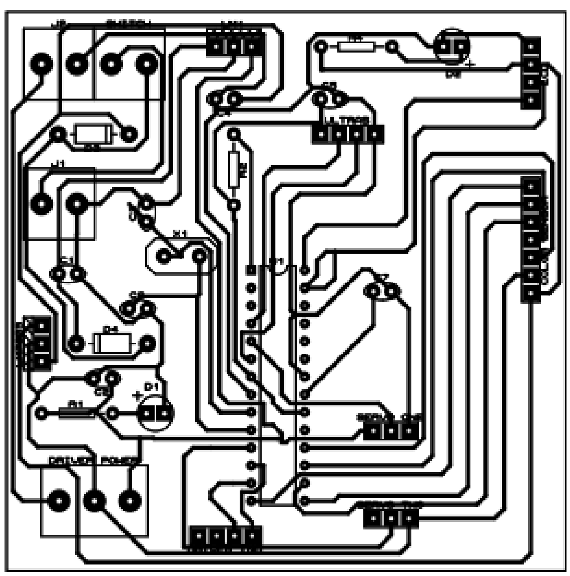

Figure 18.pcb layout

5.3 Circuit Prototype

Figure 19,circuit prototype

CHAPTER SIX
CONCLUSION AND RECOMMENDATION

6.1 Conclusion

The designing of this project was successful since the aim of this project was accomplished as the procedures and methodologies had been followed, it has been developed by integrating feature of all the hard ware components used. Presence of all module has been reasoned out and placed carefully contributing to the best working of the unit

6.2 Recommendation

I would like to recommend that, the project of Automatic Fruit Sorter and Counting Machine be developed more as technology become wide by adding or introduce backup of electrical supply system such as solar power and generator when absence of electricity, introduce other means of color detection like cameras and high efficient sensors and detecting other fruits features like ripeness and size for better sorting while still maintaining the freshness and quality of the fruits

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2. Unitec – cherry Author(s): Patel, R., Desai, P., & Shah, A. Conference: Proceedings of the International Conference on Computer Science and Technology Year: 2009.
3. Kim, H., Lee, S., & Park, J. Journal: Computers and Electronics in Agriculture Year: 2005 DOI: [DOI Number].
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APPENDICES A
Source code

```
0// Libraries #include <Wire.h> #include <Servo.h> #include <TimerOne.h>

#include <LiquidCrystal_I2C.h> // Include the LiquidCrystal_I2C library for the
LCD display


// Set the LCD address to 0x27 for a 16 characters and 2 line display

LiquidCrystal_I2C lcd(0x27, 16, 2);


// Servo object Servo servo1;


// Define color sensor pins #define S0 3

#define S1 4

#define S2 5

#define S3 6 #define ldrPin A0 #define laserPin 8

#define sensorOut 7


// Calibration Values

// Get these from Calibration Sketch

int redMin = 43;    // Red minimum value int redMax = 172;     // Red maximum
value int greenMin = 58; // Green minimum value int greenMax = 246; // Green
```

maximum value int blueMin = 48; // Blue minimum value int blueMax = 186; // Blue maximum value

// Variables for Color Pulse Width Measurements int redPW = 0;

int greenPW = 0; int bluePW = 0;

// Variables for final Color values int redValue;

int greenValue; int blueValue;

// Latching variables

bool redDetected = false, greenDetected = false, blueDetected = false;

bool latchRed = false, latchGreen = false, latchBlue = false, passCheckLatch = false;

// Time holding arrays

unsigned long arrayRed[20] = {}; unsigned long arrayGreen[20] = {}; unsigned long arrayBlue[20] = {};

// Object counting variables int redCount = 0;

int greenCount = 0; int blueCount = 0; int totalCount = 0;

// Motor Variables and libraries

// Pin definitions const int IN1 = 11; const int IN2 = 12; const int IN3 = 13;

```
const int IN4 = 10;


// Constants

const int STEPS_PER_REVOLUTION = 4096;


const byte SEQUENCE[8][4] = {


   { 1,   0,   0,   0  },
   { 1,   1,   0,   0  },
   { 0,   1,   0,   0  },
   { 0,   1,   1,   0  },
   { 0,   0,   1,   0  },
   { 0,   0,   1,   1  },
   { 0,   0,   0,   1  },
   { 1,   0,   0,   1  }
};


// Global variables volatile bool run = false; volatile int stepCount = 0;

volatile bool isClockwise = true;

volatile int stepInterval = 700; // microseconds between steps (1000us = 1ms)
```

```
//

// Set up function void setup() {

// Turn on the backlight and clear the display

// Initialize LCD lcd.begin();

lcd.backlight(); // Turn on the LCD backlight lcd.clear();  // Clear the LCD display

lcd.setCursor(0, 0); lcd.print(F(" item sorting ")); lcd.setCursor(0, 1); lcd.print(F("
and counting ")); delay(1000);


lcd.clear(); lcd.setCursor(0, 0);

lcd.print(F(" initializing ")); for (int i = 0; i < 7; i++) {

lcd.setCursor(0, 1); lcd.print(F("      sensors      ")); delay(200);

lcd.setCursor(0, 1); lcd.print(F(".....sensors   "));

delay(200);

}

lcd.setCursor(0, 1);

lcd.print(F(" sensors      DONE ")); delay(1000);


setupPins(); setupSensor();


digitalWrite(laserPin, 1);
```

```
// Setting up stepper motor initializePins();


lcd.clear(); lcd.setCursor(0, 0);

lcd.print(F("starting motors"));

for (int pos = 0; pos <= 90; pos += 1) { // goes from

0 degrees to 90 degrees

// in steps of 1 degree

servo1.write(pos); // tell servo to go to position in variable 'pos'

delay(15);    // waits 15 ms for the servo to reach the position

}

for (int pos = 90; pos >= 0; pos -= 1) { // goes from

90 degrees to 0 degrees

servo1.write(pos);  // tell servo to go to position in variable 'pos'

delay(15);    // waits 15

ms for the servo to reach the position

}

lcd.clear();


Timer1.initialize(250); // Initialize timer with 100 microsecond interval

Timer1.attachInterrupt(motorISR); Timer1.start();
```

```
for (int i = 0; i < 8; i++) { lcd.setCursor(i, 1); lcd.print(F(">")); lcd.setCursor(15 - i, 1); lcd.print(F("<")); delay(300);

}

// delay(1000);

for (int i = 0; i < 5; i++) { run = true;

delay(500); run = false; delay(100);

isClockwise = !isClockwise;

}

isClockwise = false; delay(2000);

run = true;

}

// Main loop(function caller) void loop() {

readColorValues(); determineColor(); colorTiming(); passCheck(); displayObjectCounting();

}

// This function sets up the pins for I/O void setupPins() {

// Set S0 - S3 as outputs pinMode(S0, OUTPUT); pinMode(S1, OUTPUT); pinMode(S2, OUTPUT); pinMode(S3, OUTPUT);


// Set LDR as input
```

pinMode(ldrPin, INPUT); pinMode(laserPin, OUTPUT);

// Set Sensor output as input pinMode(sensorOut, INPUT);

servo1.attach(9);

}

// This function sets sensor's sensitivity void setupSensor() {

// Set Frequency scaling to 20% digitalWrite(S0, HIGH); digitalWrite(S1, LOW);

}

// This function sets color values for red, green and blue

void readColorValues() {

// Read and map Red value

redPW = getPulseWidth(S2, LOW, S3, LOW); redValue = map(redPW, redMin, redMax, 255, 0); delay(100);

// Read and map Green value

greenPW = getPulseWidth(S2, HIGH, S3, HIGH); greenValue = map(greenPW, greenMin, greenMax, 255, 0); delay(100);

// Read and map Blue value

bluePW = getPulseWidth(S2, LOW, S3, HIGH); blueValue = map(bluePW, blueMin, blueMax, 255, 0);

```
delay(100);

}

// This function returns color values

int getPulseWidth(int S2State, int S2Val, int S3State, int S3Val) {

digitalWrite(S2State, S2Val); digitalWrite(S3State, S3Val); return
pulseIn(sensorOut, LOW);

}


// This function determines the color based on intensity of R,G and B values

void determineColor() {

if ((redValue > greenValue) && (redValue > blueValue))

{ //add this (&& redValue > 123,blueValue > 103,greenValue > 23) to ensure the
colour is well detected and set

redDetected = true; blueDetected = false; greenDetected = false;

} else if ((greenValue > blueValue) && (greenValue > redValue)) { // add this (&&
redValue > 103,blueValue > 23,greenValue > 123) to ensure the colour is well
detected and set

greenDetected = true; redDetected = false; blueDetected = false;

} else if ((blueValue > greenValue) && (blueValue > redValue)) { // add this (&&
redValue > 23,blueValue > 123,greenValue > 103) to ensure the colour is well
detected and set
```

```
blueDetected = true; redDetected = false; greenDetected = false;

} else {

redDetected = false; blueDetected = false; greenDetected = false;


latchRed = false; latchBlue = false; latchGreen = false;

}

}


// This function recoords the time the colour was detected

void colorTiming() {

// Initializing static variables static int i = 0, j = 0, k = 0;


// Increment variables i += 1;

j += 1;

k += 1;

// Reset variables before they overflow if (i >= 20) {

i = 0;

}

if (j >= 20) { j = 0;

}
```

```
if (k >= 20) { k = 0;

}


// Updating timers and values if (redDetected && !latchRed) {

redCount += 1; latchRed = true;

arrayRed[i] = millis();

} else if (greenDetected && !latchGreen) { greenCount += 1;

latchGreen = true; arrayGreen[j] = millis();

} else if (blueDetected && !latchBlue) { blueCount += 1;

latchBlue = true; arrayBlue[k] = millis();

}

totalCount = redCount + blueCount + greenCount;

}


// This function drives servo motor based on the read array

void passCheck() {

if ((analogRead(ldrPin) < 1020) && !passCheckLatch) { for (int i = 0; i < 20; i++) {

if (findLeastTimeAtIndex(i) != 0) { passCheckLatch = true; servo1.write(0);

break;

} else { servo1.write(60);
```

```
}

}

} else if ((analogRead(ldrPin) > 1020) && passCheckLatch) {

delay(1000); // Short delay for the items to fall in their respective places

passCheckLatch = false;

}

}

// Function to find which array has the least non-zero time at a specific index

int findLeastTimeAtIndex(int index) {

// Ignore zeros and compare values at the current index

if (arrayRed[index] != 0 && (arrayRed[index] < arrayGreen[index]) &&
(arrayRed[index] < arrayBlue[index])) {

arrayRed[index] = 0;

return 1;

} else if (arrayGreen[index] != 0 && (arrayGreen[index] < arrayBlue[index])) {

arrayGreen[index] = 0;

return 2;

} else if (arrayBlue[index] != 0) { arrayBlue[index] = 0;

return 3;

} else {

return 0; // All values are zero
```

```
}

}

// Motor controlling functions void initializePins() {

pinMode(IN1, OUTPUT); pinMode(IN2, OUTPUT); pinMode(IN3, OUTPUT);
pinMode(IN4, OUTPUT);

}


void motorISR() {

static unsigned long lastStepTime = 0; unsigned long currentTime = micros();

stepInterval = constrain(stepInterval, 700, 30000);


if (run) {

if (currentTime - lastStepTime >= stepInterval) { lastStepTime = currentTime;

performStep();

}

} else { stopMotor();

}

}


void performStep() {

int stepIndex = stepCount % 8; if (!isClockwise) {
```

```
stepIndex = (8 - stepIndex) % 8; // Reverse the sequence for counter-clockwise

}

oneStep(stepIndex); stepCount++;

if (stepCount >= STEPS_PER_REVOLUTION) {

stepCount = 0;

}

}

void oneStep(int step) { digitalWrite(IN1, SEQUENCE[step][0]); digitalWrite(IN2,
SEQUENCE[step][1]); digitalWrite(IN3, SEQUENCE[step][2]); digitalWrite(IN4,
SEQUENCE[step][3]);

}

void stopMotor() { digitalWrite(IN1, LOW); digitalWrite(IN2, LOW);
digitalWrite(IN3, LOW); digitalWrite(IN4, LOW);

}

void setDirection(bool clockwise) { isClockwise = clockwise;

}

// LCD functions

void displayObjectCounting() { lcd.clear(); lcd.setCursor(0, 0);

lcd.print(F("R    G    B    CNT ")); lcd.setCursor(0, 1); lcd.print(redCount);
lcd.setCursor(4, 1); lcd.print(greenCount); lcd.setCursor(8, 1);
```

APPENDIX B

The time frame work plan of the project

activities	nov	dec	jan	feb	much	april	may	Jun	july
Title selection and defending									
Supervisor consultation									
literature review									
Data collection									
Data analysis									
Circuit designing									
Circuit implementation and testing									

COST ESTIMATION

The following are the all cost which used so as to accomplish this project

S/N	Name	Quantity/Materials	Cost@	Cost
	ARDUINO ON PCB COMPONENTS			
1	ATmega328p Microcontroller IC	1	10000	10000/=
2	28 Pin IC Holder	1	1000	1000/=
3	16 MHz Quartz Crystal	1	1000	1000/=
4	22pf capacitor	2	500	1000/=
5	Resistors – 470 ohms	6	200	1200/=
6	LEDs x 2 (Red and Green)	1 yellow,1green	200	400/=
7	Push Button	2	300	600/=
8	5V Regulator (7805)	1	2000	2000/=
10	1N4007 Diode	3	300	900/=
11	10uf and	3	500	1500/=
12	Screw Terminal (Terminal block)	2	1000	2000/=
13	Male Headers	2	1000	2000/=
14	Female Headers	2	1000	2000/=
15	100nf capacitor	2	200	400/=
16	Resistors – 10k	3	200	600/=
17	Power switch	1	1000	1000/=
18	3.7 volt batteries	3	5000	15000/=
	PROJECT BASED COMPONENTS			
1	Connecting wires (Ethernet)	1 meter	1500	1500/=

2	Screws (for joints and few fastening)		4000	4000/=
3	Stepper motor	1	15000	15000/=
4	Servo motor	1	15000	15000/=
5	Ultrasonic sensor	1	10000	10000/=
6	Wood glue***	1	7000	7000/=
7	Housing (dimensions will Wooden case be given)		40000	4000/=
	OTHERS			
1	Hot glue	1 stick	1000	1000/=
2	Rubber bands	20	50	1000/=
3	Black spray	1	5000	5000/=
4	Solder wire	1m	1000	1000/=
5	PCB design costs (etching Fe3Cl, PCB, Gloss paper and printing)		5500	5500/=
	TOTAL			112,600/=

Table 10 Cost estimation of the project

APPENDIX C

QUESTIONNAIRE ABOUT AUTOMATIC FRUITS SORTER AND COUNTING MACHINE

1. What is the primary function of an automatic fruit sorter?

A) To wash fruits

B) To sort fruits by size, color, or quality

C) To package fruits

D) To transport fruits

2. Which technology is commonly used in automatic fruit sorting systems?

A) GPS

B) Barcode scanning

C) Optical sensors

D) Radio frequency identification (RFID)

3. What is the main advantage of using a counting machine for fruits?

A) It increases the taste of the fruit

B) It reduces manual labor and increases accuracy

C) It improves the color of the fruit

D) It enhances the nutritional value of the fruit

4. Which fruit characteristic is typically detected by optical sensors in a sorting machine?

A) Weight

B) Texture

C) Color

D) Sweetness

5. What is the typical speed range of an automatic fruit sorter?

A) 1-5 fruits per minute

B) 10-50 fruits per minute

C) 100-500 fruits per minute

D) 1000-5000 fruits per minute

6. Which of the following fruits are commonly sorted by automatic systems?

A) Apples and oranges

B) Mangoes and pineapples

C) Berries and grapes

D) All of the above

7. What is a key benefit of sorting fruits by size and color?

A) It reduces the fruit’s sugar content

B) It enhances the aesthetic appeal and market value

C) It increases the fruit’s shelf life

D) It lowers the production cost

8. What type of data do counting machines typically provide?

A) Fruit weight only

B) Count of fruits and batch information

C) Nutritional data of fruits

D) None of the above

9. Which material is commonly used in the construction of fruit sorting belts?

A) Steel

B) Plastic

C) Glass

D) Aluminum

10. What is the main function of the sorting algorithm in a fruit sorting machine?

A) To increase the size of the fruits

B) To differentiate fruits based on specific criteria

C) To make the fruits taste better

D) To cool the fruits

11. Which industry benefit most from the use of fruit sorting and counting machines?

A) Entertainment industry

B) Agriculture and food processing

C) Construction

D) Textile manufacturing

12. How do fruit sorting machines improve efficiency in packing?

A) By reducing the fruit’s weight

B) By automatically arranging fruits into pre-defined packages

C) By increasing the fruit’s ripening speed

D) By decreasing the size of the fruits

13. Which component of a fruit sorter is crucial for accurate sorting by color?

A) Conveyor belt

B) Optical camera

C) Sorting bin

D) Motor

14. What is a major challenge in the development of advanced fruit sorting machines?

A) Increasing the fruit’s sweetness

B) Handling fruits with varying shapes and sizes

C) Reducing the fruit’s ripening time

D) Enhancing the fruit's natural flavor

15. Which of the following is an essential feature of modern counting machines for fruits?

A) Manual operation

B) Touchscreen interface

C) Limited capacity

D) High power consumption